

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Additional guidance                                                                                                                                                                                                                                                                                      | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 7(a)            | <p>An explanation that includes four of the following points:</p> <ul style="list-style-type: none"> <li>• reference to negative feedback (1)</li> <li>• thermoreceptors detect {increase / change} in temperature / temperature above 37°C} (1)</li> <li>• impulses to {thermoregulatory centre / hypothalamus / heat loss centre} (1)</li> <li>• (thermoregulatory centre) sends impulses {to correctly named effector / via motor neurones / via sympathetic neurones} (1)</li> <li>• correct response by named effectors to {increase heat loss / decrease heat energy gain} from the body (1)</li> </ul> | <p>ignore signals / messages</p> <p>accept receptors detecting temperature change</p> <p>ignore control centre unqualified</p> <p>e.g. sweat glands, hair erector muscles, liver, smooth muscles in skin blood vessels</p> <p>e.g. vasodilation, sweat production, decreased / change metabolic rate</p> | (4)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Additional guidance                                                                                                                                                                                                                         | Mark |
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| 7(b)            | <p>An answer that includes three of the following points:</p> <ul style="list-style-type: none"> <li>the loop of Henle is (very) long / more juxtamedullary nephrons (1)</li> <li>(more) sodium ions pumped out of the ascending limb / {lower water potential / higher solute concentration} of tissue fluid (1)</li> <li>more water reabsorbed results in concentrated urine production (1)</li> <li>accept correct behavioural adaptation (1)</li> </ul> | <p>accept more counter current multiplication</p> <p>accept kangaroo rats have more {aquaporins / ADH}</p> <p>accept increased permeability of {descending limb / collecting duct}</p> <p>e.g. nocturnal / stays underground during day</p> | (3)  |

| Question number | Answer                                                                                                                                                                                                                           | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                       | Mark |
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| 7(c)(i)         | <p>A calculation showing the following steps:</p> <ul style="list-style-type: none"> <li>calculation of weekly urine produced (1)</li> <li>calculation of difference and answer given in standard form with units (1)</li> </ul> | <p><math>(0.45 \times 7) = 3.15</math> AND<br/> <math>(1500 \times 7) = 10500</math></p> <p><math>(10.500 - 3.15) = 10496.85 \text{ cm}^3</math><br/> <math>= 1.0497 \times 10^4 \text{ cm}^3</math><br/> accept <math>1.05 \times 10^4 \text{ cm}^3</math> / <math>1.0 \times 10^4 \text{ cm}^3</math><br/> <math>10496.9 = 1</math> mark or <math>10.5 \text{ dm}^3</math><br/> no units max 1 mark</p> | (2)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                   | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 7(c)(ii)        | <p>An answer that includes four of the following points</p> <ul style="list-style-type: none"> <li>• kangaroo rat has a lower water balance (1)</li> <li>•</li> <li>• a correct difference / comparison (1)</li> </ul><br><ul style="list-style-type: none"> <li>• with an appropriate linked explanation (1)</li> </ul><br><ul style="list-style-type: none"> <li>• a second correct difference / comparison (1)</li> </ul><br><ul style="list-style-type: none"> <li>• with an appropriate linked explanation (1)</li> </ul> | <p>accept converse for human<br/>accept in both water gain = water loss per day<br/>e.g. more water is gained (as a percentage) through metabolic reactions in kangaroo rat (than human)</p><br><p>because kangaroo rat drinks no liquid / lives in desert</p><br><p>e.g. kangaroo rat produces very little urine</p><br><p>as it has a very long loop of Henle / produces concentrated urine / rat lives in {dry place / desert}</p> | (4)  |